Crop Recommendation Report

Lucknow, Uttar Pradesh • 2023

Top Recommended Crop	Sugarcane	Confidence: 27.3%
District	Lucknow	Uttar Pradesh
Assessment Year	2023	Crops Evaluated: 8

Recommendation Explanation

Sugarcane is the top recommendation for Lucknow in 2023. It has consistently delivered strong yields in this district, with a historical average of 68645 kg/ha and a previous-year yield of 66161 kg/ha. Its stable cultivation history makes it a dependable choice under current climate conditions.

Full Crop Ranking

Rank	Crop	Confidence (%)	Prev. Yield (kg/ha)	Hist. Mean (kg/ha)	Area Share (%)
1	**Sugarcane**	**27.3%**	**66161**	**68645**	**23.5%**
2	Wheat	14.3%	3714	3792	9.1%
3	Rice	11.2%	3076	2978	8.9%
4	Lentil	10.4%	1097	928	8.0%
5	Maize	9.8%	3248	3011	10.1%
6	Chickpea	9.7%	1055	1154	16.9%
7	Mustard	9.5%	1445	1300	6.6%
8	Groundnut	7.8%	1608	1584	16.8%

Key Influencing Factors

Factor	Influence Direction	Magnitude
Monsoon Season Precipitation	↑ Favourable	Medium
Growing Season Temperature	→ Neutral	Low
Previous Year Yield vs Historical Average	↑ Favourable	Low
District Area Share	→ Neutral	Low
Irrigation Coverage Ratio	↑ Favourable	High

Climate Summary

Metric	Value
Annual Precipitation	990 mm
Monsoon Season Precipitation	848 mm
Average Maximum Temperature	32.0 °C

Average Minimum Temperature	18.1 °C
Irrigation Coverage Ratio	82.20%

This report was generated by the AgriCAST AI Crop Recommendation System using a LambdaMART Learning-to-Rank model trained on historical agricultural and climate data. Recommendations are decision-support tools and should be verified with local agronomists before operational use. AgriCAST v1.0